Pseudo-Random Functions and Authenticated Encryption

Palash Sarkar

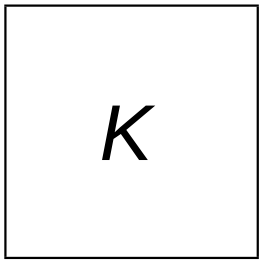
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16 December 2010

Presented at the
C. R. Rao Advanced Institute of
Mathematics, Statistics and Computer Science



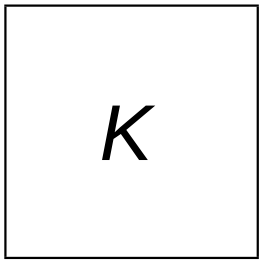
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- K is a secret 128-bit value.
- K is uniformly distributed over $\{0,1\}^{128}$.



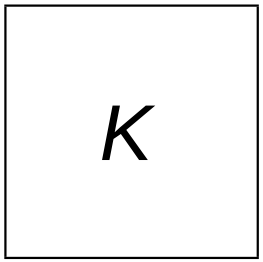
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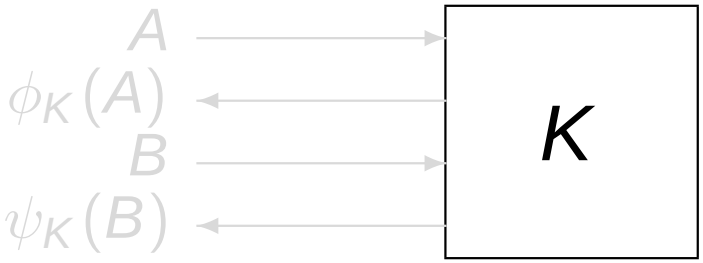
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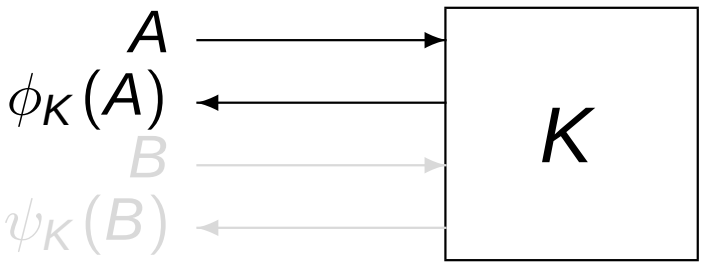
Using a Secret



- Cryptographic methods interact with the environment using K.
- The interaction is via certain functions (ϕ and ψ).
- The values $\phi_K(A)$ or $\psi_K(B)$ are public.
- Each such value potentially leaks information about K.
- How to argue about security?



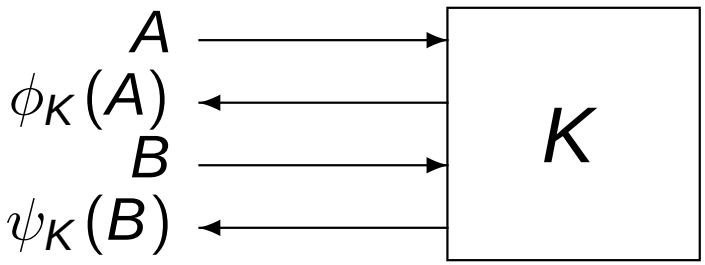
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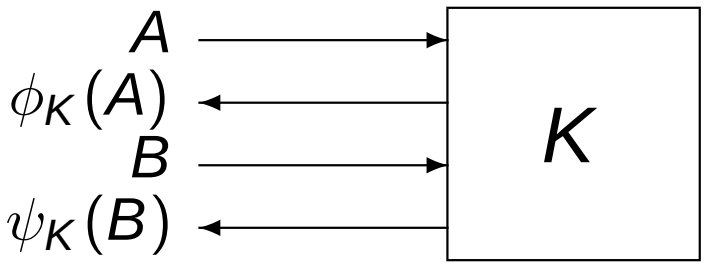
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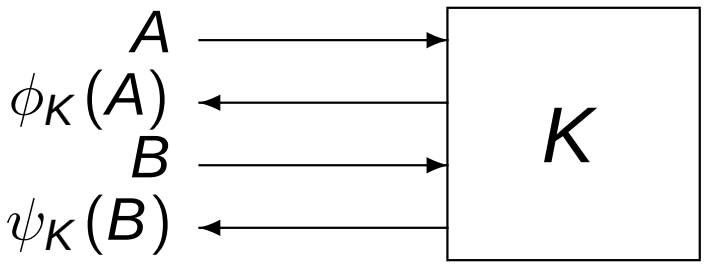
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Functions Parametrized by a Secret

$$\phi : \mathcal{K} \times \mathcal{D} \rightarrow \mathcal{R}; \qquad \phi_K(A) \stackrel{\Delta}{=} \phi(K, A).$$

- An adversary can see the values

$$V_1 = \phi_K(A_1), V_2 = \phi_K(A_2), V_3 = \phi_K(A_3), \ldots.$$

- An adversary may choose the values $A_1, A_2, A_3, \ldots$.
- Idea of security: The values $V_1, V_2, V_3, \ldots$ should appear independent and uniform to the adversary.
- How to formalise this notion?



Functions Parametrized by a Secret (contd.)

$$\phi : \mathcal{K} \times \mathcal{D} \rightarrow \mathcal{R}; \qquad \phi_K(A) \stackrel{\Delta}{=} \phi(K, A).$$

- $\mathcal{K}$, $\mathcal{D}$ and $\mathcal{R}$ are finite sets.
- Let $\mathcal{K} = \{0,1\}^k$ and so there are 2^k possible functions ϕ.
- Secret $K \equiv$ Uniform Random K
  $\Rightarrow \phi$ is specified by k bits of randomness.

Random function $\phi : \mathcal{D} \rightarrow \mathcal{R}$.

- ϕ is chosen uniformly at random from the set of possible 2^k functions indexed by $\mathcal{K}$.



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Uniform Random Function

$$f : \mathcal{D} \rightarrow \mathcal{R}.$$

- There are a total of $|\mathcal{R}|^{|\mathcal{D}|}$ functions from $\mathcal{D}$ to $\mathcal{R}$.
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A useful view

if $A_1, A_2, A_3, \ldots$ are *distinct* values
then $f(A_1), f(A_2), f(A_3)$ are independent and uniformly distributed.



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Random Versus Uniform Random

- ϕ is a random function: k bits of randomness.
- f is a uniform random function: $|\mathcal{D}| \log |\mathcal{R}|$ bits of randomness.
- A typical set of values: $k = 128$; $|\mathcal{D}| = 2^{128}$; $|\mathcal{R}| = 2^{128}$.
- Gap in randomness: 128 versus 128×2^{128}.
- If the adversary can perform any amount of computation, then this gap in randomness can be detected.
- To obtain a meaningful definition, we restrict the adversary to be computationally bounded.



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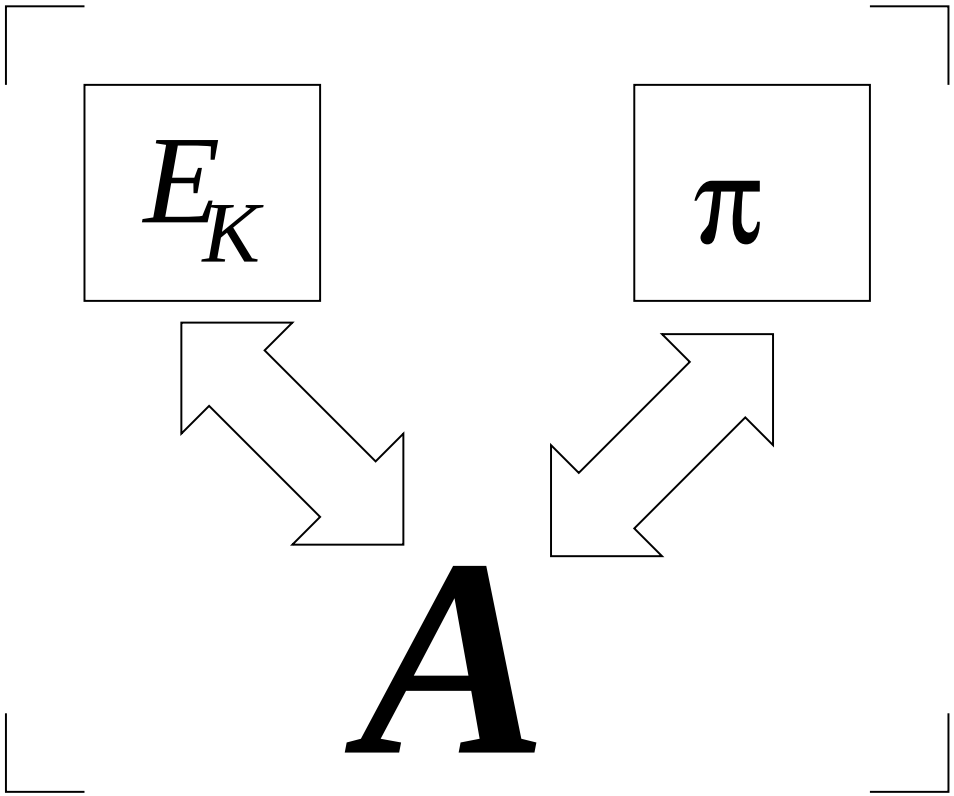


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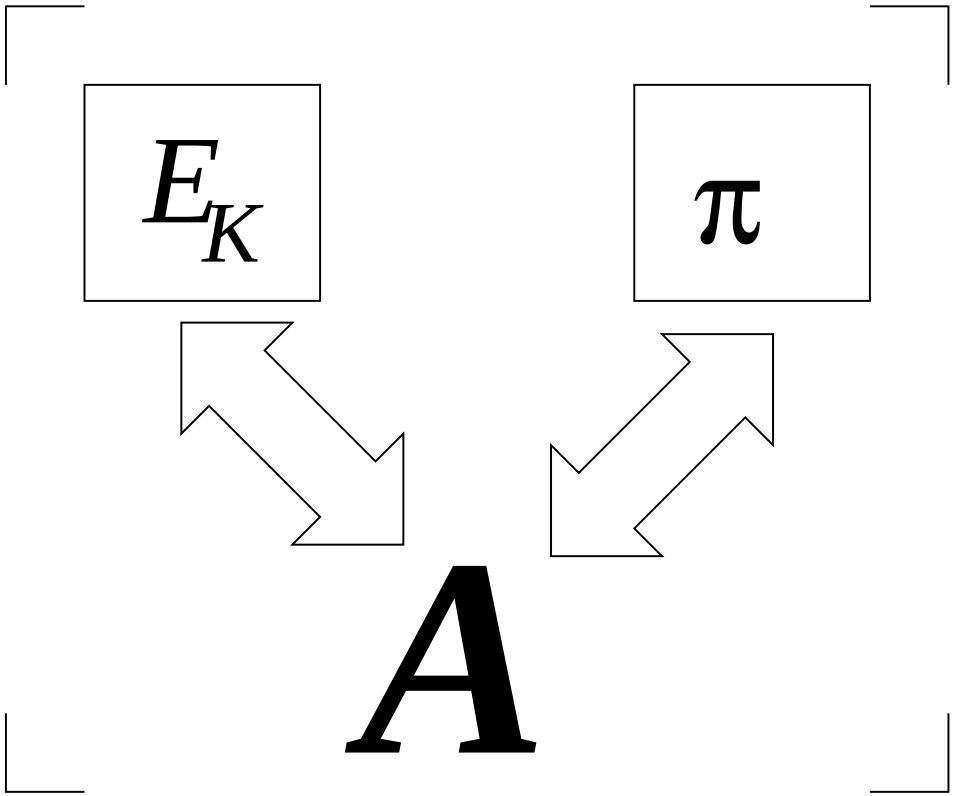


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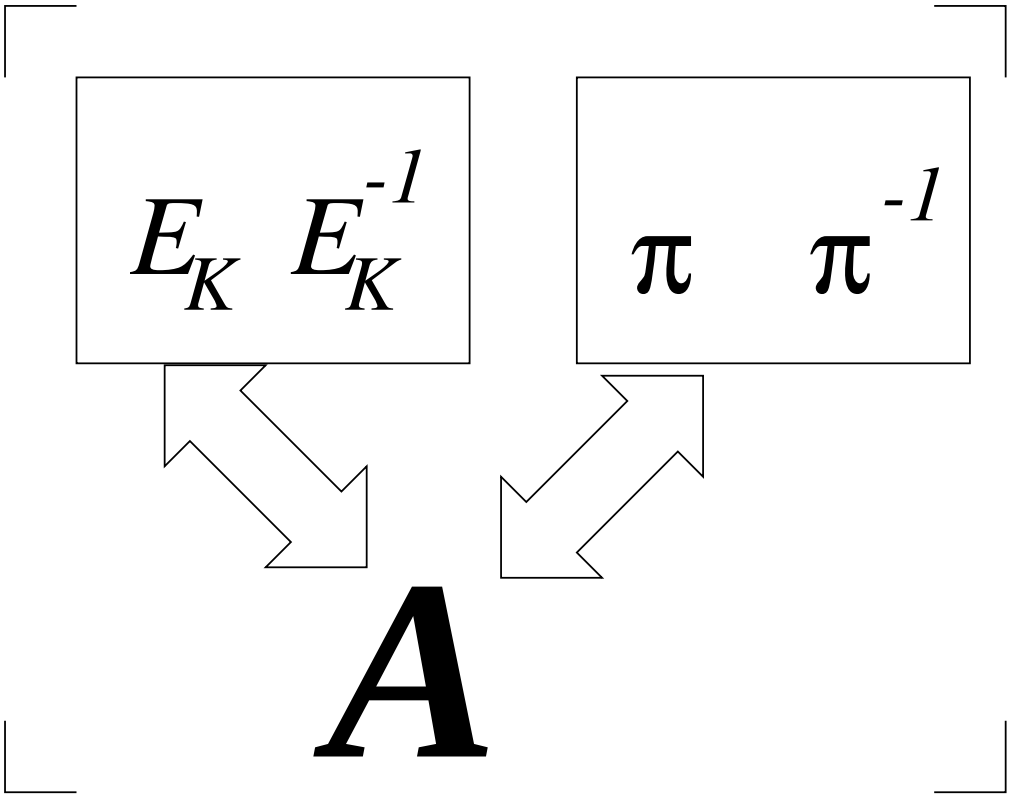
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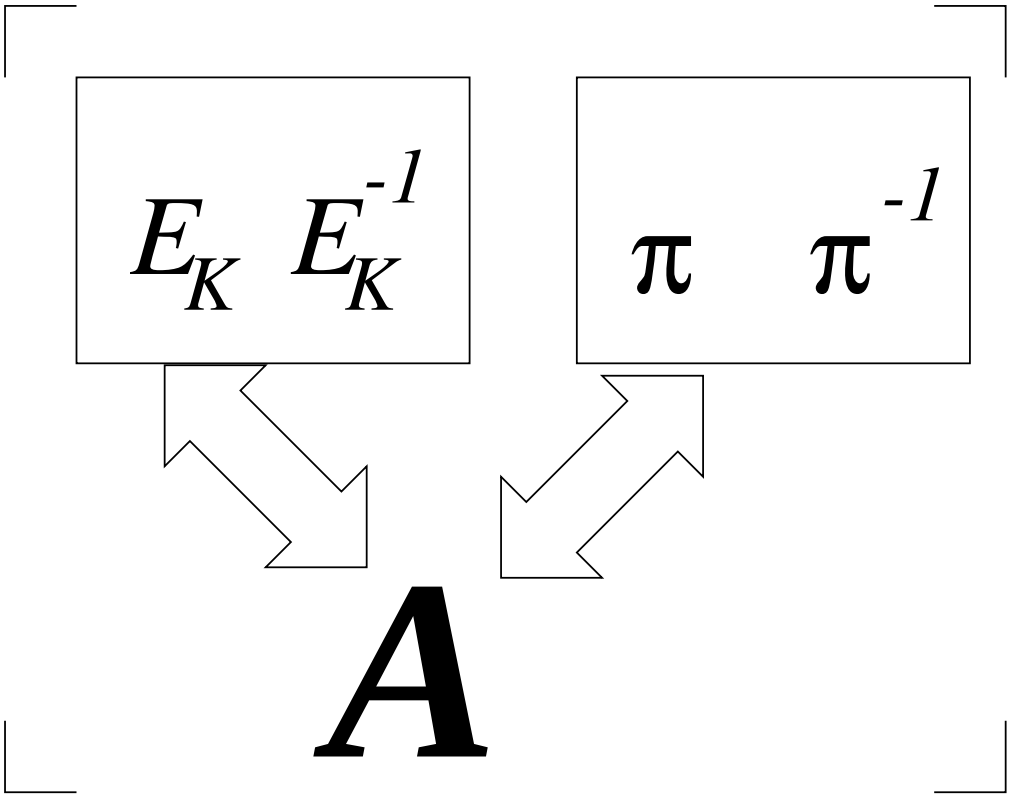
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Ideal Block Cipher

- Assumption: either a PRP or an SPRP.
  - PRP is enough to show confidentiality.
  - SPRP required to show authentication.
- With the above assumption, we allow the adversary to be computationally unbounded.

So what else is there to do?



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Block Cipher and Practical Requirements

Block length.

- The block cipher handles n-bit blocks.
- Typically, $n = 128, 192$ or 256 bits.

Message Requirements.

- Handle “long” messages.
- Handle “variable” length messages.



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- Authenticated encryption.
- Authenticated encryption with associated data.
- Others.



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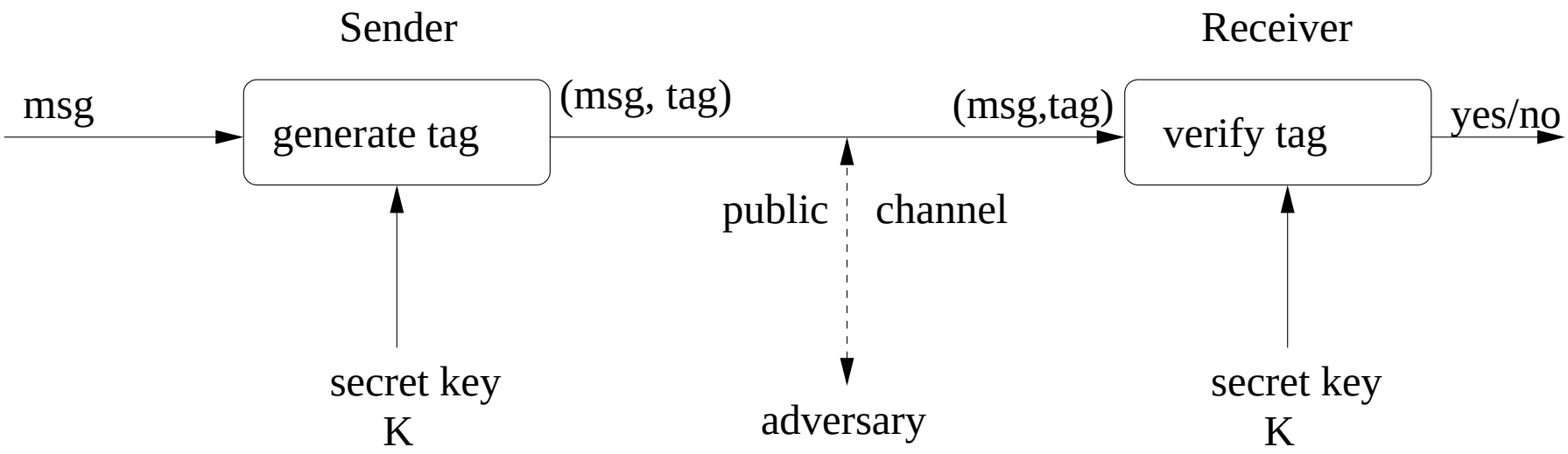
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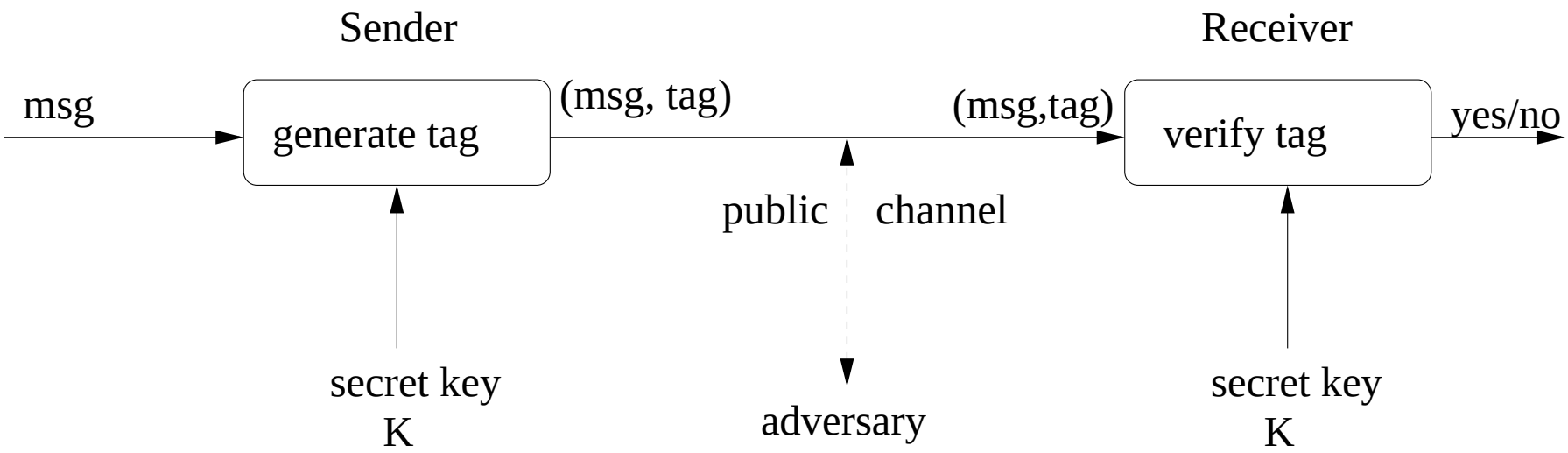


Authentication: General Idea



- Adversary chooses messages $M_1, M_2, M_3, \ldots$.
- Recieves corresponding tags $\text{tag}_1, \text{tag}_2, \text{tag}_3, \ldots$.
- Adversary produces (M, tag).
- Success: if (M, tag) passes verification.

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Authentication: Arguing Security

Let

$$f : \text{messages} \rightarrow \text{tags}.$$

- f is built using a block cipher E_K.
- E_K is modelled as an SPRP π.
- So, f is a random function;
  but, not a uniform random function.



Authentication: Arguing Security

$$f : \text{messages} \rightarrow \text{tags}.$$

- If f is a PRF, then authentication is achieved.
- If a t-bit tag is produced, then

$$\mathbf{Adv}_f^{\text{auth}}(q, \sigma) = \frac{1}{2^t} + \mathbf{Adv}_f^{\text{prf}}(q, \sigma).$$

q: number of queries made by the adversary;
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Domain Extenders

- Block cipher based construction of PRF.
- Secret key of the block cipher is the key of the PRF.
- Let $\rho : \mathcal{U} \rightarrow \mathcal{Y}$ be a random function. A function $f : \mathcal{X} \rightarrow \mathcal{Y}$ is said to be a domain extender for ρ if

$$f = \rho \circ f_1^{(\rho)}$$

  where $f_1 : \mathcal{X} \rightarrow \mathcal{U}$ and
  - On any input, f_1 invokes ρ a finite number of times.
  - The only randomness of f_1 comes from the invocations of ρ.
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Collision Probabilities

$$f : \mathcal{X} \rightarrow \mathcal{Y}.$$

Let x_1, x_2 be two *distinct* elements of $\mathcal{X}$.

$$\Pr[f(x_1) = f(x_2)]$$

is called a collision probability for f.

- ε-CR:
  if all collision probabilities are at most ε.
- ε-CR with respect to λ:
  if all collision probabilities are at most $\varepsilon \times \max(\lambda(x_1), \lambda(x_2))$.

($\lambda(x) = \#$ n-bit blocks in x.)



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Domain Extenders: Some Crucial Notions

$$f = \rho \circ f_1.$$

Distinct x, x' with $\lambda(x) = m + 1$, $\lambda(x') = m' + 1$. Define

- $Z = f_1(x)$, $Z' = f_1(x')$.
- $U_1, \ldots, U_m$: inputs to the different invocations of ρ in the computation of $f_1(x)$.
  $U'_1, \ldots, U'_m$ similarly defined from $f_1(x')$.
- Self-Disjoint(x): $\bigwedge_{i=1}^{m} (Z \neq U_i)$.
- Pairwise-Disjoint(x, x'): $\left(\bigwedge_{i=1}^{m} (Z' \neq U_i) \wedge \bigwedge_{j=1}^{m'} (Z \neq U'_j) \right)$.



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Domain Extenders: Some Crucial Notions

$$f = \rho \circ f_1.$$

f_1 is $(\varepsilon_1, \varepsilon_2)$-disjoint with respect to λ, if for all pairs of distinct $x, x' \in \mathcal{X}$,

- $\Pr[\overline{\text{Self-Disjoint}(x)}] \leq \varepsilon_1(\lambda(x))$.
- $\Pr[\overline{\text{Pairwise-Disjoint}(x, x')}] \leq \varepsilon_2(\lambda(x) + \lambda(x'))$.



Domain Extenders: Security Theorem

$$f = \rho \circ f_1.$$

Suppose that

- f_1 is ε-CR with respect to the length function λ;
- f_1 is $(\varepsilon_1, \varepsilon_2)$-disjoint with respect to λ.

Then

- If ρ is a uniform random function, then $\mathbf{Adv}_f(q, \sigma) \leq \sigma(q\varepsilon + \varepsilon_1 + 2q\varepsilon_2)$.
- If ρ is a uniform random permutation, then $\mathbf{Adv}_f(q, \sigma) \leq \sigma(q\varepsilon + \varepsilon_1 + 2q\varepsilon_2) + \frac{q\sigma}{\#\mathcal{Y}}$.



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Domain Extenders: Security Theorem

$$f = \rho \circ f_1.$$

- Reduces the task of proving PRF property of f to that of computing certain probablities for f_1.
  - Collision probabilities.
  - Self disjoint probabilities.
  - Pairwise disjoint probabilities.
- Reduces q queries to 2 queries.
- Probabilities computed over the randomness of ρ.
  - We will focus on the case where ρ is a uniform random permutation.
  - Recall that a uniform random permutation is an idealization of a block cipher.



A sketch of a construction



Masking Functions

$\psi : \mathbb{F}_{2^n} \rightarrow \mathbb{F}_{2^n}$ be linear and $\phi_\beta(i) = \psi^i(\beta)$.

ϕ is a proper masking function if

- $\Pr[\phi_\beta(k) = \alpha] = 1/2^n$.
- $\Pr[\phi_\beta(k_1) \oplus \phi_\beta(k_2) = \alpha] = 1/2^n$.
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for any α; non-negative integer k; integers k_1, k_2; uniform random β; uniform random (β_1, β_2) chosen without replacement;
Instantiations.

- Powering-up: $\phi : i \mapsto u^i \beta \bmod \tau(u)$.
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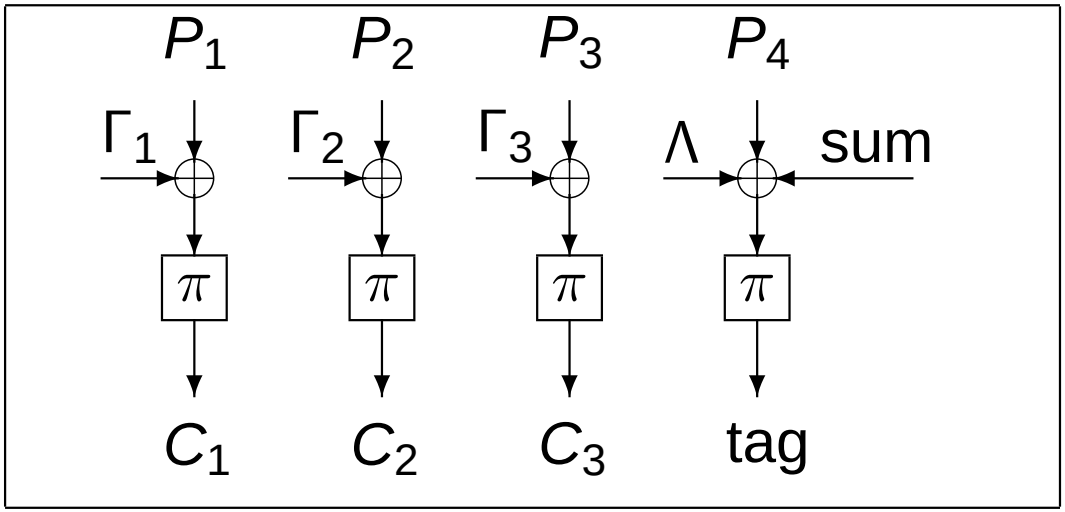
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iPMAC: A Parallelizable PRF



- $\gamma = \pi(\text{str})$; str is a fixed string.
- $\Gamma_i = \phi_\gamma(i)$;
- $\Lambda = 0^n$ if the last block is full and $\Lambda = \phi_\gamma(4)$ if the last block has been padded;
- sum $= C_1 \oplus C_2 \oplus C_3$.



Security of iPMAC

$$\text{iPMAC} = \pi \circ \text{iPHASH}$$

iPHASH.

- produces the input just before the final π invocation.
- collision probabilities: at most $\max(m, m')/2^n$;
- self-disjoint probabilities: at most $2m/2^n$;
- pairwise-disjoint probabilities: at most $(m + m')/2^n$;

Theorem.

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Efficiency of iPMAC

The ‘i’ in iPMAC **denotes improvement over** PMAC **(Rogaway)**

- Easily reconfigurable: simply changing the field representation changes the algorithm; PMAC requires a design stage discrete log computation.
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Authenticated Encryption



Formal Definitions

$$f : \mathcal{N} \times \mathcal{X} \rightarrow \mathcal{X} \times \{0,1\}^n.$$

For $f(N,X) = (Y, \text{tag})$, $\text{len}(X) = \text{len}(Y)$.

- $f^{\text{main}}(N,X) = Y$ if $f(N,X) = (Y, \text{tag})$.
- f is an AE-function if for every $N \in \mathcal{N}$, $f_N^{\text{main}}(\cdot) \triangleq f^{\text{main}}(N,\cdot)$ is a length preserving permutation.
  The invertibility of f_N^{main} ensures that decryption is possible, i.e., given Y, it is possible to obtain X.
- For an AE-function f, $\widetilde{f}(N,Y) = \text{tag}$ if $f(N,X) = (Y, \text{tag})$ for some $X \in \mathcal{X}$; $\widetilde{f}$: authentication function associated with f.

$\mathcal{F}_n[\mathcal{N}, \mathcal{X}]$ be the set of all such functions.



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Let f be a random AE-function and f^* be a function distributed uniformly over $\mathcal{F}_n[\mathcal{N}, \mathcal{X}]$.

- Adversary $\mathcal{A}$ has oracle access to f: $\mathcal{A}$ can adaptively query f on $(N^{(s)}, P^{(s)})$ and get back $(C^{(s)}, \mathrm{tag}^{(s)})$ in return.
- Nonce-respecting adversary: the nonces of two different queries cannot be equal. Such an adversary is called nonce-respecting.
- Finally, $\mathcal{A}$ outputs a bit.

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AE: Defining Authentication

- Adversary $\mathcal{A}$ has oracle access to f: queries $(N^{(s)}, X^{(s)})$ are made adaptively with the restriction that the nonces are distinct.
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- Finally, $\mathcal{A}$ outputs a forgery $(N^{(q)}, Y^{(q)}, \mathrm{tag}^{(q)})$.
- Restriction: $(N^{(q)}, Y^{(q)}, \mathrm{tag}^{(q)}) \neq (N^{(s)}, Y^{(s)}, \mathrm{tag}^{(s)})$ for all $1 \leq s \leq q-1$.
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Success: if there is an $X^{(q)}$ such that $f(N^{(q)}, X^{(q)}) = (Y^{(q)}, \mathrm{tag}^{(q)})$.

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AE-function $f(N, X) = (Y, \text{tag})$.

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Showing authentication property of an AE-function can be reduced to

- Showing privacy of f^{main}.
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PAE: Sarkar 2010

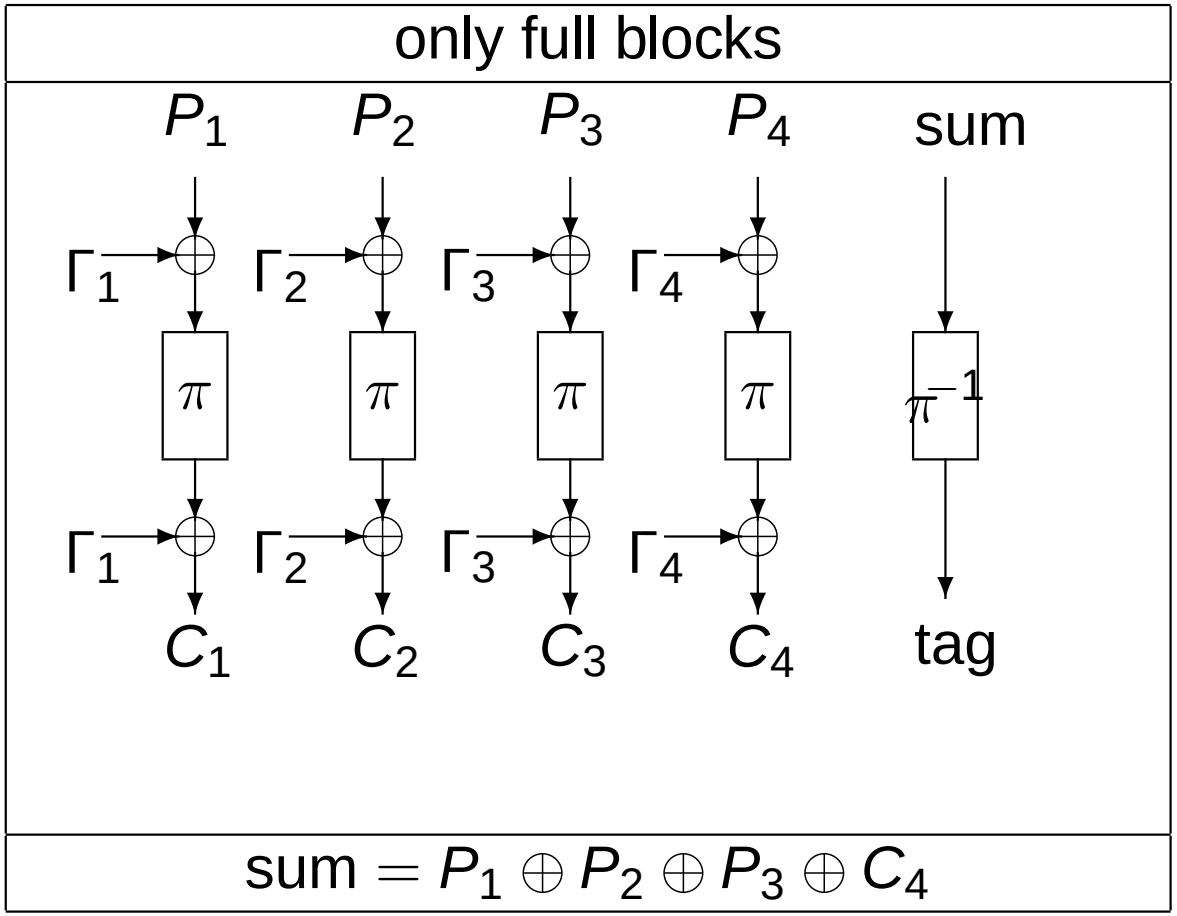


Figure: Encryption using PAE: $\gamma = \pi^{-1}(N)$; $\Gamma_i = \phi_\gamma(i)$.



PAE: Sarkar 2010 (Contd.)

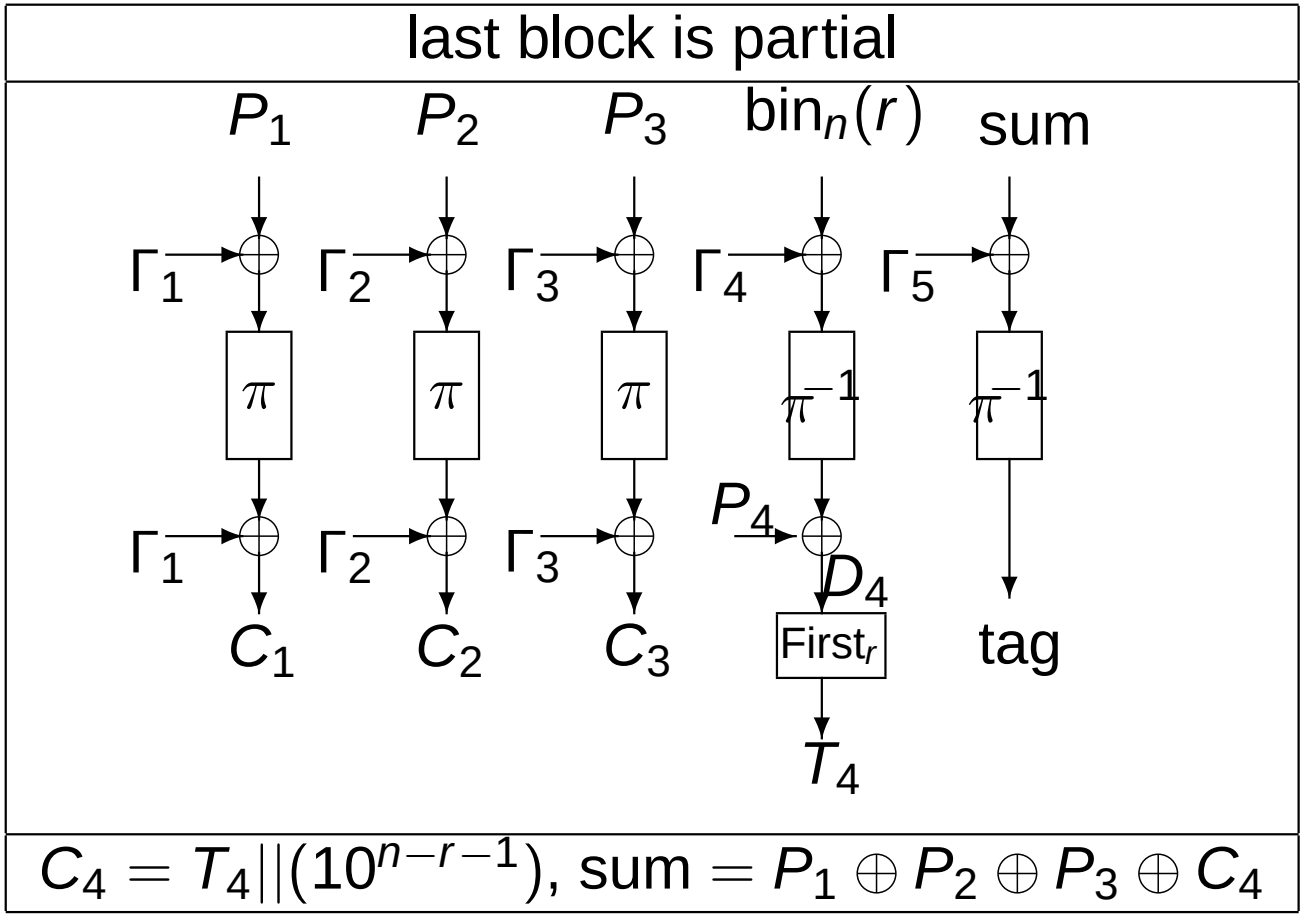


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Theorem (Privacy).

$$\mathbf{Adv}^{\mathrm{priv}}_{\mathrm{PAE}}(q, \sigma) \leq \frac{2(\sigma + 2q)^2}{2^n}.$$

Consequently,

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- $\widetilde{\text{PAE}}_\pi$ is equal to $\text{miPMAC}_{\pi^{-1}}$.
- $\text{miPMAC}_{\pi^{-1}}$ is a modified version of iPMAC and has the same security bound.
- PAE was designed so as to achieve this equality.

For every $N \in \{0,1\}^n$ and binary string C, with $\text{len}(C) \geq 0$,

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$$\mathbf{Adv}^{\text{aeauth}}_{t\text{-PAE}}(q, \sigma) \leq \frac{1}{2^t} + \frac{2(\sigma+2q)^2}{2^n} + \frac{\sigma(7q+2)}{2^n}.$$



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$$\mathbf{Adv}^{\text{aeauth}}_{t\text{-PAE}}(q, \sigma) \leq \frac{1}{2^t} + \frac{2(\sigma+2q)^2}{2^n} + \frac{\sigma(7q+2)}{2^n}.$$



Authentication of PAE

- $\widetilde{\text{PAE}}_\pi$ is equal to $\text{miPMAC}_{\pi^{-1}}$.
- $\text{miPMAC}_{\pi^{-1}}$ is a modified version of iPMAC and has the same security bound.
- PAE was designed so as to achieve this equality.

For every $N \in \{0,1\}^n$ and binary string C, with $\text{len}(C) \geq 0$,

$$\widetilde{\text{PAE}}_\pi(N, C) = \text{miPMAC}_{\pi^{-1}}(N, C).$$

Consequently,

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Conclusion

- Secrecy, randomness and cryptography.
- Pseudo-random permutation and functions.
- Arguing about the security of authentication and authenticated encryption schemes.
- A parallel authentication scheme.
- A parallel AE scheme.
- Brief idea of security.



Thank you for your attention!

